TOPIC 8.3
Electric Fields

Required Course Content

LEARNING OBJECTIVE

8.3.A
Describe the electric field produced by a charged object or configuration of point charges.

ESSENTIAL KNOWLEDGE

8.3.A.1
Electric fields may originate from charged objects.

8.3.A.2
The electric field at a given point is the ratio of the electric force exerted on a test charge at the point to the charge of the test charge.

Relevant equation:

$$\vec{E} = \frac{\vec{F}_E}{q}$$

8.3.A.2.i
A test charge is a point charge of small enough magnitude such that its presence does not significantly affect an electric field in its vicinity.

8.3.A.2.ii
An electric field points away from isolated positive charges and toward isolated negative charges.

8.3.A.2.iii
The electric force exerted on a positive test charge by an electric field is in the same direction as the electric field.

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SUGGESTED SKILLS

1.B
Create quantitative graphs with appropriate scales and units, including plotting data.

2.A
Derive a symbolic expression from known quantities by selecting and following a logical mathematical pathway.

2.D
Predict new values or factors of change of physical quantities using functional dependence between variables.

3.A
Create experimental procedures that are appropriate for a given scientific question.

3.B
Apply an appropriate law, definition, theoretical relationship, or model to make a claim.

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Electric Charges, Fields, and Gauss's Law

LEARNING OBJECTIVE	ESSENTIAL KNOWLEDGE
8.3.A Describe the electric field produced by a charged object or configuration of point charges.	8.3.A.3 The electric field is a vector quantity and can be represented in space using vector field maps. 8.3.A.3.i The net electric field at a given location is the vector sum of individual electric fields created by nearby charged objects. 8.3.A.3.ii Electric field maps use vectors to depict the magnitude and direction of the electric field at many locations within a given region. 8.3.A.3.iii Electric field line diagrams are simplified models of electric field maps and can be used to determine the relative magnitude and direction of the electric field at any position in the diagram.
8.3.B Describe the electric field generated by charged conductors or insulators.	8.3.B.1 While in electrostatic equilibrium, the excess charge of a conductor is distributed on the surface of the conductor, and the electric field within the conductor is zero. 8.3.B.1.i At the surface of a charged conductor, the electric field is perpendicular to the surface. 8.3.B.1.ii The electric field outside an isolated sphere with spherically symmetric charge distribution is the same as the electric field due to a point charge with the same net charge as the sphere located at the center of the sphere. 8.3.B.2 While in electrostatic equilibrium, the excess charge of an insulator is distributed throughout the interior of the insulator as well as at the surface, and the electric field within the insulator may have a nonzero value.

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